

# Auction with Secret Reserve Price

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# 1 Model Outline

- One seller,  $N$  bidders.
- Seller's cost is either  $c_L$  or  $c_H$  with  $\Pr(c_L) = \psi$ .
- Buyer's value is i.i.d. distributed according to some  $F$  supported on  $[c_L, 1]$ , with density  $f$ .
- **First stage:** an English auction with initially hidden reserve price, which is revealed conditional on being met.
- **Second stage:** if the English auction fails, one bidder is randomly drawn to make a TIOLI (take-it-or-leave-it) offer to the seller. An individual bidder is drawn with probability  $\delta \leq 1/N$ .

**Notation.**  $\underline{r}$  and  $\bar{r}$ : endogenous lower and upper support of the low-cost seller's mixed strategy (chosen in equilibrium, not drawn);  $r_H$ : high-cost seller's reserve;  $\rho \in \Delta[\underline{r}, \bar{r}]$ : cdf of the reserve  $r$  actually played on the path;  $\beta$ : symmetric stage-1 bidding strategy;  $\tilde{\psi}$ : belief that the seller is low-cost.

## 2 Preliminary Analysis

### 2.1 Second-stage belief and offer

If the chosen buyer has a belief  $\tilde{\psi}$ , he offers price  $c_H$  if and only if

$$v - c_H \geq \tilde{\psi}(v - c_L) \iff v \geq \frac{c_H - \tilde{\psi}c_L}{1 - \tilde{\psi}}. \quad (1)$$

### 2.2 Equilibrium structure

Consider an equilibrium where the high-cost seller plays a pure strategy  $r_H$  and the low-cost seller plays a mixed strategy denoted by  $\rho \in \Delta[\underline{r}, \bar{r}]$ .

**Claim 1.**  $r_H > \bar{r}$ .

Proof in Section 6. For any increasing bidding function, the optimal reserve price for a high-cost seller is strictly above that of a low-cost seller.

### 2.3 High-cost seller

When the bid exceeds  $\bar{r}$  but the reserve is not met, bidders believe the seller is high-cost and bid according to  $\beta(v) = v - \delta(v - c_H)$ . Once a bidder has  $\beta(v) \geq r_H$ , the reserve is announced as met and people bid their own values. The high-cost seller's profit  $\Pi_H(r_H)$  is

$$\Pi_H(r_H) = \int_{\beta^{-1}(r_H)}^1 \int_{c_L}^v (\max\{r_H, s\} - c_H) d\left(\frac{F(s)}{F(v)}\right)^{N-1} dF(v)^N. \quad (2)$$

Simplifying,

$$\begin{aligned}
\Pi_H(r_H) &= N \int_{\beta^{-1}(r_H)}^1 \left[ (r_H - c_H) F(r_H)^{N-1} + \int_{r_H}^v (s - c_H) dF(s)^{N-1} \right] dF(v) \\
&= N \left[ (r_H - c_H) F(r_H)^{N-1} (1 - F(\beta^{-1}(r_H))) + \int_{r_H}^1 (s - c_H) dF(s)^{N-1} \right. \\
&\quad \left. - F(\beta^{-1}(r_H)) \int_{r_H}^{\beta^{-1}(r_H)} (s - c_H) dF(s)^{N-1} - \int_{\beta^{-1}(r_H)}^1 F(v)(v - c_H) dF(v)^{N-1} \right].
\end{aligned} \tag{3}$$

The first-order condition is

$$r_H - c_H = (1 - \delta) \frac{1 - F\left(\frac{r_H - \delta c_H}{1 - \delta}\right)}{f\left(\frac{r_H - \delta c_H}{1 - \delta}\right)} - \frac{1}{F(r_H)^{N-1}} \int_{r_H}^{(r_H - \delta c_H)/(1 - \delta)} (v - c_H) dF(v)^{N-1}. \tag{4}$$

## 2.4 Belief after a failed English auction

If the English auction fails with final bid  $b$ , the buyer's belief is

$$\tilde{\psi} = \frac{\psi(1 - \rho(b))}{1 - \psi\rho(b)}. \tag{5}$$

## 3 Bidding in the First-Stage English Auction

Suppose a bidder with value  $v$  decides his bid by comparing  $v - b$  against

$$\delta \max \left\{ v - c_H, \frac{\psi(1 - \rho(b))}{1 - \psi\rho(b)} (v - c_L), 0 \right\}. \tag{6}$$

His bid thus satisfies

$$b = v - \delta \max \left\{ v - c_H, \frac{\psi(1 - \rho(b))}{1 - \psi\rho(b)} (v - c_L), 0 \right\}. \tag{7}$$

The right-hand side is weakly decreasing in  $b$  and bounded between  $[v - \delta \max\{v - c_H, \psi(v - c_L)\}, v - \delta \max\{v - c_H, 0\}]$ . The bidding strategy  $\beta$  given  $\rho$  is thus

$$\beta(v) = \begin{cases} v - \delta \psi(v - c_L) & \text{if } v \leq \frac{r - \delta \psi c_L}{1 - \delta \psi}, \\ h^{-1}(v) & \text{if } v \in \left( \frac{r - \delta \psi c_L}{1 - \delta \psi}, v^* \right), \\ v - \delta(v - c_H) & \text{if } v \geq v^*, \end{cases} \tag{8}$$

where

$$h(b) \equiv \frac{(1 - \psi\rho(b))b - \delta\psi(1 - \rho(b))c_L}{1 - \delta\psi - (1 - \delta)\psi\rho(b)}, \tag{9}$$

i.e.  $h(b)$  solves  $v - b = \delta\tilde{\psi}(b)(v - c_L)$  with  $\tilde{\psi}(b) = \psi(1 - \rho(b))/(1 - \psi\rho(b))$  from (5), and  $v^* \in [(\underline{r} - \delta\psi c_L)/(1 - \delta\psi), (c_H - \psi c_L)/(1 - \psi)]$  is the solution to

$$v - c_H = \frac{\psi(1 - \rho((1 - \delta)v + \delta c_H))}{1 - \psi\rho((1 - \delta)v + \delta c_H)}(v - c_L) \iff \rho((1 - \delta)v + \delta c_H) = \frac{\psi(v - c_L) - (v - c_H)}{\psi(c_H - c_L)}. \quad (10)$$

## 4 Low-Cost Seller Profit

It suffices to focus on the low-type seller's profit. Suppose it sets reserve price  $\underline{r}$ .

### 4.1 When the auction fails

The English auction *fails* when every bidder stops before the reserve  $\underline{r}$  is met, so the sale moves to stage 2. On the low bidding branch, failure is equivalent to  $v_{\max} \leq z$  with

$$z \equiv \beta^{-1}(\underline{r}) = \frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}. \quad (11)$$

This event has probability  $\Pr(\text{fail}) = F(z)^N$ .

The seller may still earn positive profit in stage 2 if the selected buyer makes a  $c_H$  offer; each accepted  $c_H$  offer yields profit  $c_H - c_L$ . When the auction fails before  $\underline{r}$ , this event reveals no information about the seller's type, so the posterior belief that the seller is low-cost remains  $\psi$ . By (1) with this posterior, a buyer with value  $v$  offers  $c_H$  in stage 2 if and only if  $v \geq v_\psi$ , where

$$v_\psi \equiv \frac{c_H - \psi c_L}{1 - \psi}. \quad (12)$$

The subscript records that here  $\tilde{\psi} = \psi$ . Since failure implies  $v_{\max} \leq z$ , the seller earns this margin only when

$$v_\psi < v_{\max} < z \quad (\text{equivalently } z > v_{\max} > v_\psi).$$

Let  $\underline{r}_0$  denote the reserve at which  $z = v_\psi$ ,

$$\underline{r}_0 \equiv \delta\psi c_L + (1 - \delta\psi) \frac{c_H - \psi c_L}{1 - \psi} = \frac{(1 - \delta\psi)c_H - \psi(1 - \delta)c_L}{1 - \psi}. \quad (13)$$

Since  $z$  is increasing in  $\underline{r}$ ,  $v_\psi < v_{\max} < z$  is possible if and only if  $\underline{r} > \underline{r}_0$ ; otherwise  $\Pi_{\text{fail}} = 0$ .<sup>1</sup>

For  $\underline{r} > \underline{r}_0$ , one bidder is drawn to make a take-it-or-leave-it offer with probability  $\delta$ . Here we distinguish between the highest bidder and the others, because stage 1 bidding reveals  $v_{\max}$ , which the seller observes, but not the other bidders' values.

If the selected bidder is the highest, his value equals  $v_{\max}$ , which the seller already knows, and if  $v_{\max} \geq v_\psi$  he offers  $c_H$ . This contributes

$$\Pi_{\text{fail}}^{(1)} = \delta(c_H - c_L) \int_{v_\psi}^z dF(v)^N. \quad (14)$$

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<sup>1</sup>The cutoff  $\underline{r}_0$  need not lie below the top of the value support. If buyers place sufficiently high prior probability on the low-cost type,  $v_\psi$  is so large that even the largest feasible  $v_{\max}$  on a failure path need not warrant a  $c_H$  offer. Indeed,  $\underline{r}_0 < 1$  if and only if  $\psi < (1 - c_H)/(1 - c_L - \delta(c_H - c_L))$ ; if not, then  $\underline{r}_0 \geq 1$  and  $\Pi_{\text{fail}} = 0$  for every feasible reserve  $\underline{r} \leq 1$ .

If the selected bidder is not the highest, there are  $N-1$  symmetric possibilities. The seller still knows  $v_{\max}$  but not the selected value  $v$ , and a non-max bidder with value  $v$  offers  $c_H$  with probability  $1 - F(v_\psi)/F(v)$ . This contributes

$$\Pi_{\text{fail}}^{(2)} = \delta(N-1)(c_H - c_L) \int_{v_\psi}^z \left(1 - \frac{F(v_\psi)}{F(v)}\right) dF(v)^N. \quad (15)$$

Adding (14) and (15), and factoring out  $\delta(c_H - c_L)$ ,

$$\Pi_{\text{fail}} = \Pi_{\text{fail}}^{(1)} + \Pi_{\text{fail}}^{(2)} = \delta(c_H - c_L) \left[ \int_{v_\psi}^z dF(v)^N + (N-1) \int_{v_\psi}^z \left(1 - \frac{F(v_\psi)}{F(v)}\right) dF(v)^N \right]. \quad (16)$$

The first integral in (16) equals  $F(z)^N - F(v_\psi)^N$ . For  $N \geq 2$ , integration by parts on the second integral gives

$$\int_{v_\psi}^z \left(1 - \frac{F(v_\psi)}{F(v)}\right) dF(v)^N = \left(1 - \frac{F(v_\psi)}{F(z)}\right) F(z)^N - \frac{F(v_\psi)}{N-1} (F(z)^{N-1} - F(v_\psi)^{N-1}), \quad (17)$$

since  $\int_{v_\psi}^z F(v)^{N-2} f(v) dv = (F(z)^{N-1} - F(v_\psi)^{N-1})/(N-1)$ . Adding the two integrals in (16),

$$\begin{aligned} & \int_{v_\psi}^z dF(v)^N + (N-1) \int_{v_\psi}^z \left(1 - \frac{F(v_\psi)}{F(v)}\right) dF(v)^N \\ &= F(z)^N - F(v_\psi)^N + (N-1) \left(1 - \frac{F(v_\psi)}{F(z)}\right) F(z)^N - F(v_\psi) (F(z)^{N-1} - F(v_\psi)^{N-1}) \\ &= N(F(z) - F(v_\psi)) F(z)^{N-1}, \end{aligned} \quad (18)$$

where the second line uses  $F(v_\psi)^N = F(v_\psi) F(v_\psi)^{N-1}$ . When  $N = 1$ , only the first integral enters and (18) still holds. Substituting into (16),

$$\Pi_{\text{fail}} = \delta N (c_H - c_L) (F(z) - F(v_\psi)) F(z)^{N-1}. \quad (19)$$

Raising  $\underline{r}$  makes stage 1 failure more likely because  $F(z)^N$  increases, and for  $\underline{r} > \underline{r}_0$  it also enlarges  $[v_\psi, z]$  and raises the non-max offer probability  $1 - F(v_\psi)/F(v)$ .

## 4.2 When the auction succeeds

The auction *succeeds* when the reserve is met in stage 1, so the sale does not move to stage 2. This is equivalent to  $v_{\max} > z$ , with probability  $1 - F(z)^N$ . Once the reserve is met, bidders bid truthfully in the English auction. The highest-value bidder  $v$  wins and pays  $\max\{\underline{r}, s\}$ , so the seller's margin is  $\max\{\underline{r}, s\} - c_L$ . Averaging over the winner's value  $v \geq z$  and the runner-up value  $s \leq v$  gives

$$\Pi_{\text{succ}} = \int_z^1 \int_{c_L}^v (\max\{\underline{r}, s\} - c_L) d\left(\frac{F(s)}{F(v)}\right)^{N-1} dF(v)^N, \quad (20)$$

where  $d(F(s)/F(v))^{N-1}$  is the distribution of  $s$  conditional on  $v_{\max} = v$ . Integrating by parts yields the equivalent form

$$\Pi_{\text{succ}} = N \left\{ (\underline{r} - c_L) F(\underline{r})^{N-1} (1 - F(z)) + \int_z^1 \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \right\}. \quad (21)$$

The first term collects states where the runner-up's value is below  $\underline{r}$ , so the sale price equals the reserve. The double integral collects states where  $s > \underline{r}$ , so the winner pays the runner-up's value  $s$  (the usual second-price payment in an English auction).

### 4.3 Total profit and $\underline{r}^*$

Total profit is

$$\Pi(\underline{r}) = \mathbf{1}\{\underline{r} > \underline{r}_0\} \delta N(c_H - c_L) (F(z) - F(v_\psi)) F(z)^{N-1} + \Pi_{\text{succ}}, \quad (22)$$

The indicator  $\mathbf{1}\{\underline{r} > \underline{r}_0\}$  makes explicit the reserve cutoff below which stage-2 profit is zero. Stage 2 offers always use the buyers' posterior belief  $\tilde{\psi}$ . In (22), failure before  $\underline{r}$  leaves  $\tilde{\psi} = \psi$ , so the cutoff is  $v_\psi$ . If the seller deviates upward, stage-2 payoffs use the updated posterior  $\tilde{\psi}$  from (5), which is strictly below  $\psi$ .

**Claim 2.** *The equilibrium lower support  $\underline{r}^*$  is the argmax of  $\Pi(\cdot)$  in (22).*

### 4.4 Proof of Claim 2

*Proof.* Let  $\underline{r}^* \in \arg \max \Pi(\cdot)$ . We show that any other lower support  $\underline{r} \neq \underline{r}^*$  cannot be optimal.

*Downward deviation.* If  $\underline{r} > \underline{r}^*$ , failure still occurs before the announced reserve, so the posterior remains  $\tilde{\psi} = \psi$  and (22) is exact, so  $\Pi(\underline{r}^*) > \Pi(\underline{r})$ .

*Upward deviation.* If  $\underline{r} < \underline{r}^*$ , announce  $\underline{r}^*$  and write  $z \equiv \beta^{-1}(\underline{r})$  and  $z^* \equiv \beta^{-1}(\underline{r}^*)$ .

*Case 1:*  $\underline{r}^* > \underline{r} > \underline{r}_0$ . Then  $z > v_\psi$  and  $z^* > z$ . True failure profit is

$$\begin{aligned} \Pi_{\text{fail}}^{\text{true}} = & \mathbf{1}\{\underline{r} > \underline{r}_0\} \delta N(c_H - c_L) (F(z) - F(v_\psi)) F(z)^{N-1} \\ & + \delta(c_H - c_L) \int_z^{z^*} dF(v)^N + \delta(N-1)(c_H - c_L) \int_z^{z^*} \left(1 - \frac{F(v_\rho(v))}{F(v)}\right) dF(v)^N, \end{aligned} \quad (23)$$

where the first term is  $\Pi_{\text{fail}}$  and  $v_\rho(v)$  is the  $c_H$ -offer cutoff with belief  $\tilde{\psi}(\beta(v))$  on  $(z, z^*]$ . On  $(z, z^*]$ ,  $\tilde{\psi}(\beta(v)) < \psi$ , so  $v_\rho(v) < v_\psi < z < z^*$  and the last two terms are strictly positive. The failure term in  $\Pi(\underline{r}^*)$  uses  $v_\psi$  up to  $z^*$ , so the updated-belief block strictly exceeds its contribution. Success profit at  $\underline{r}^*$  is unchanged, hence  $\Pi_{\text{fail}}^{\text{true}} > \Pi(\underline{r}^*) > \Pi(\underline{r})$ .

*Case 2:*  $\underline{r}^* > \underline{r}$  but  $\underline{r} \leq \underline{r}_0$ . Then  $\mathbf{1}\{\underline{r} > \underline{r}_0\} = 0$  and  $\Pi_{\text{fail}} = 0$ . If  $\underline{r}^* > \underline{r}_0$ , the integrals over  $(z, z^*]$  in (23) are positive; otherwise compare success profit only.

*Conclusion.* Either deviation is profitable, so  $\underline{r}^*$  is the optimal lower support.  $\square$

The low-cost seller's equilibrium payoff is  $\Pi(\underline{r}^*)$ . On the low branch,  $\beta^{-1}(\underline{r}) = (\underline{r} - \delta\psi c_L)/(1 - \delta\psi)$ .

## 5 Low-Cost Seller: Optimal Support and Mixed Strategy

This section completes the low-cost seller's problem. Claim 2 pins the equilibrium lower support to the reserve that maximizes  $\Pi(\cdot)$  in (22). We characterize that maximizer, extend payoffs to reserves above the lower support on the mixing path, and construct the reserve distribution from the resulting indifference condition.

### 5.1 Optimal lower support

The lower support  $\underline{r}^*$  maximizes  $\Pi(\underline{r})$  in (22). For an interior maximizer with  $\underline{r}^* > \underline{r}_0$  from (13),  $\mathbf{1}\{\underline{r}^* > \underline{r}_0\} = 1$  in (22). Imposing  $d\Pi(\underline{r})/d\underline{r} = 0$  and rearranging yields the first-order condition

$$\begin{aligned} \underline{r}^* - c_L &= (1 - \delta\psi) \frac{1 - F(\beta^{-1}(\underline{r}^*))}{f(\beta^{-1}(\underline{r}^*))} \\ &+ \frac{1}{F(\underline{r}^*)^{N-1}} \left[ \delta(c_H - c_L) F(\beta^{-1}(\underline{r}^*))^{N-2} \left( NF(\beta^{-1}(\underline{r}^*)) - (N-1)F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right) \right. \\ &\quad \left. - \int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)} (s - c_L) dF(s)^{N-1} \right]. \end{aligned} \quad (24)$$

The derivation is in Appendix A.

### 5.2 Profit and indifference on the mixing path

In equilibrium the low-cost seller mixes a reserve  $r \in [\underline{r}, \bar{r}]$  with cdf  $\rho$ , where  $\underline{r}$  and  $\bar{r}$  are the endogenous bounds of the support. If the realized reserve exceeds  $\underline{r}$ , stage 1 failure can occur with  $v_{\max}$  between  $\beta^{-1}(\underline{r})$  and  $\beta^{-1}(r)$ ; bids then reveal  $\rho(\beta(v)) > 0$  and beliefs update via (5). On the failure path, beliefs split at  $\beta^{-1}(\underline{r})$ : the prior  $\psi$  applies below that threshold and  $\tilde{\psi}(\beta(v))$  applies above it until  $\beta^{-1}(r)$ . Payoff from announcing  $r > \underline{r}$  is

$$\begin{aligned} \hat{\Pi}(r) &= \delta N(c_H - c_L) \left[ \left( F(\beta^{-1}(\underline{r})) - F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right) F(\beta^{-1}(\underline{r}))^{N-1} \right. \\ &\quad \left. + F(\beta^{-1}(r))^N - F(\beta^{-1}(\underline{r}))^N - N \int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)} F\left(\frac{(1 - \psi\rho(\beta(v)))c_H - \psi(1 - \rho(\beta(v)))c_L}{1 - \psi}\right) \right. \\ &\quad \left. \times F(v)^{N-2} dF(v) \right] \\ &+ N \left\{ (r - c_L) F(r)^{N-1} (1 - F(\beta^{-1}(r))) \right. \\ &\quad \left. + \int_{\beta^{-1}(r)}^1 \int_r^v (s - c_L) dF(s)^{N-1} dF(v) \right\}. \end{aligned} \quad (25)$$

Appendix B derives the middle integral. When  $r = \underline{r}$ , the middle terms vanish and (25) coincides with (22).

On the equilibrium path  $\underline{r} = \underline{r}^*$ , so the seller is indifferent across the support:

$$\hat{\Pi}(r) = \Pi(\underline{r}^*) \quad \forall r \in [\underline{r}, \bar{r}]. \quad (26)$$

### 5.3 Construction of the mixing distribution

Differentiating (26) on the low branch yields an ordinary differential equation for  $h(r) \equiv \beta^{-1}(r)$  and the mixing cdf  $\rho(r)$ ; the algebra is in Appendix C. On the low branch,  $h(r)$  coincides with (9) at  $b = r$ :

$$h(r) = \frac{(1 - \psi\rho(r))r - \delta\psi(1 - \rho(r))c_L}{1 - \delta\psi - (1 - \delta)\psi\rho(r)}, \quad (27)$$

with inverse

$$\rho(r) = \frac{-h(r) + r + \delta\psi(h(r) - c_L)}{\psi(-h(r) + r + \delta(h(r) - c_L))}. \quad (28)$$

Indifference on the low branch is equivalent to

$$h'(r) = \frac{F(r)^{N-1}(1 - F(h(r)))}{f(h(r))} \times \mathcal{B}(r, h(r))^{-1}, \quad (29)$$

where

$$\begin{aligned} \mathcal{B}(r, h) &= (h - c_L)F(h)^{N-1} - \int_r^h F(v)^{N-1} dv \\ &\quad - \delta N(c_H - c_L) \left[ F(h) - F\left(\frac{(1 - \psi\rho(r))c_H - \psi(1 - \rho(r))c_L}{1 - \psi}\right) \right] F(h)^{N-2}. \end{aligned} \quad (30)$$

**Claim 3.** Suppose  $F$  and  $f$  are continuous on  $[c_L, 1]$  with  $f > 0$ . Let  $\underline{r}^*$  satisfy (24), and set

$$h^* = \frac{\underline{r}^* - \delta\psi c_L}{1 - \delta\psi},$$

the inverse of the first bidding branch in (8). Then (29) admits a local solution  $r \mapsto h(r)$  through  $(\underline{r}^*, h^*)$ .

*Proof.* At the lower end of the support,  $\rho(\underline{r}^*) = 0$ . Hence buyers use the prior belief  $\psi$ , bid according to  $v - \delta\psi(v - c_L)$ , and the inverse bid is the displayed  $h^*$ . Appendix C.3 shows that the denominator  $\mathcal{B}(\underline{r}^*, h^*)$  is strictly positive. The right-hand side of (29) is therefore continuous around  $(\underline{r}^*, h^*)$ . Peano's theorem then gives a local solution.  $\square$

A local solution is enough to define a mixed strategy on an interval  $[\underline{r}^*, r_1]$ : given  $h$  solving (29) on that interval, (28) pins down  $\rho$ , and (26) holds by construction on the low branch. For  $r > \beta(v^*)$ ,  $\beta^{-1}(r)$  is known in closed form and (47) in Appendix C extends  $\rho$  on the upper part of  $[\underline{r}, \bar{r}]$ . Together with  $\bar{r} < r_H$  (Claim 1), this yields a mixed strategy equilibrium for the low-cost seller with payoff  $\Pi(\underline{r}^*)$ .

## 6 Upper Support and $\bar{r}$

At the top of the low-cost mixing support,  $\rho(\bar{r}) = 1$ . By (5), a stage 1 failure at bid  $\bar{r}$  therefore leaves buyers with probability zero that the seller is low-cost:  $\tilde{\psi}(\bar{r}) = 0$ . By (1), a selected buyer offers  $c_H$  only if  $v \geq c_H$ .

### 6.1 Payoff at $r = \bar{r}$

Set  $\underline{r} = \underline{r}^*$  on the equilibrium path and  $r = \bar{r}$  in (25). Write  $z^* \equiv \beta^{-1}(\underline{r}^*)$  and  $z_{\bar{r}} \equiv \beta^{-1}(\bar{r})$ . On the upper part of the support,

$$z_{\bar{r}} = \frac{\bar{r} - \delta c_H}{1 - \delta} \quad \text{when } \bar{r} > \beta(v^*), \quad (31)$$

as in (8) and Appendix C.4. Then

$$\begin{aligned} \hat{\Pi}(\bar{r}) &= \delta N(c_H - c_L) \left[ \left( F(z^*) - F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right) F(z^*)^{N-1} \right. \\ &\quad \left. + F(z_{\bar{r}})^N - F(z^*)^N - N \int_{z^*}^{z_{\bar{r}}} F\left(\frac{(1 - \psi\rho(\beta(v)))c_H - \psi(1 - \rho(\beta(v)))c_L}{1 - \psi}\right) F(v)^{N-2} dF(v) \right] \\ &\quad + N \left\{ (\bar{r} - c_L) F(\bar{r})^{N-1} (1 - F(z_{\bar{r}})) \right. \\ &\quad \left. + \int_{z_{\bar{r}}}^1 \int_{\bar{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \right\}. \end{aligned} \quad (32)$$



This is (25) evaluated at the upper support. The  $\psi$  block is the same as in (22) because it depends only on  $\underline{r}^*$ , not on  $\bar{r}$ . The remaining terms are the incremental failure and success payoffs from raising the reserve from  $\underline{r}^*$  to  $\bar{r}$ .

At the upper endpoint of the failure integral,  $\rho(\beta(z_{\bar{r}})) = \rho(\bar{r}) = 1$ , so the integrand cutoff equals  $c_H$ :

$$\frac{(1 - \psi\rho(\bar{r}))c_H - \psi(1 - \rho(\bar{r}))c_L}{1 - \psi} = c_H.$$

On the equilibrium path, indifference gives

$$\hat{\Pi}(\bar{r}) = \Pi(\underline{r}^*) = \hat{\Pi}(\underline{r}^*), \quad (33)$$

where the second equality uses (25) at  $r = \underline{r}^* = \underline{r}$ .

## 6.2 Proof of Claim 1

*Proof.* The low-cost seller mixes on  $[\underline{r}^*, \bar{r}]$  with  $\rho(\bar{r}) = 1$ . At bid  $\bar{r}$ , (5) gives  $\tilde{\psi}(\bar{r}) = 0$ , so buyers assign probability zero to the low type after failure at the top of the support. If  $\bar{r} \geq r_H$ , the high-cost seller's optimal pure reserve would lie inside (or below) the low-cost mixing support.

Consider the high-cost seller's problem in (3). Its optimal reserve  $r_H$  satisfies (4) and is strictly larger than the reserve that maximizes the low-cost *pure*-strategy profit  $\Pi(\cdot)$  in (22): a high-cost seller faces a higher stage 2 cutoff and a steeper stage 1 bidding schedule  $\beta(v) = v - \delta(v - c_H)$  than the low-cost branch  $\beta(v) = v - \delta\psi(v - c_L)$  with  $\psi < 1$ . Hence  $r_H > \arg \max_r \Pi(r) \geq \underline{r}^*$ .

On the equilibrium path, the low-cost seller is indifferent across  $[\underline{r}^*, \bar{r}]$ , so  $\hat{\Pi}(\bar{r}) = \Pi(\underline{r}^*)$  by (33). The payoff  $\hat{\Pi}(\bar{r})$  in (32) is built from the low-cost bidding rule up to  $z_{\bar{r}}$  in (31). If  $\bar{r} \geq r_H$ , the high-cost seller could meet reserve  $r_H$  while bidders above  $\bar{r}$  already treat the seller as high-cost (Section 2), so the high-cost type's optimal separation reserve would not exceed the top of the low-cost support. That contradicts  $r_H > \arg \max \Pi(\cdot)$  and the definition of  $\bar{r}$  as the upper endpoint of the low-cost mixing support with  $\rho(\bar{r}) = 1$ .

Therefore  $\bar{r} < r_H$ . □

## 7 Case of Public Reserve Price

If the equilibrium is separating, the low-cost seller sets a reserve price given by

$$r_L - c_L = (1 - \delta) \frac{1 - F\left(\frac{r_L - \delta c_L}{1 - \delta}\right)}{f\left(\frac{r_L - \delta c_L}{1 - \delta}\right)} - \frac{1}{F(r_L)^{N-1}} \int_{r_L}^{(r_L - \delta c_L)/(1 - \delta)} (v - c_L) dF(v)^{N-1}. \quad (34)$$

whereas the high-cost seller sets reserve price weakly above  $r_H$ , solution to

$$r_H - c_H = (1 - \delta) \frac{1 - F\left(\frac{r_H - \delta c_H}{1 - \delta}\right)}{f\left(\frac{r_H - \delta c_H}{1 - \delta}\right)} - \frac{1}{F(r_H)^{N-1}} \int_{r_H}^{(r_H - \delta c_H)/(1 - \delta)} (v - c_H) dF(v)^{N-1}, \quad (35)$$

that is enough to deter low-type mimicking.

## A FOC for $\underline{r}^*$

This appendix derives (24) from (22) on the interior region where  $\underline{r} > \underline{r}_0$  in (22).

### A.1 Derivative of the failure-region term

$$\begin{aligned}
& \frac{d}{d\underline{r}} \left[ \delta N(c_H - c_L) \times \left( F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) - F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right) F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^{N-1} \right] \\
&= \frac{1}{1 - \delta\psi} f\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^{N-1} \\
&\quad + (N-1) \frac{1}{1 - \delta\psi} \left( F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) - F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right) F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^{N-2} f\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) \\
&= \frac{1}{1 - \delta\psi} f\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^{N-2} \left[ NF\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) - (N-1)F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right]. \quad (36)
\end{aligned}$$

### A.2 Simplifying the double integral

$$\begin{aligned}
& \int_{\beta^{-1}(\underline{r})}^1 \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \\
&= \left[ F(v) \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} \right]_{\beta^{-1}(\underline{r})}^1 - \int_{\beta^{-1}(\underline{r})}^1 F(v)(v - c_L) dF(v)^{N-1} \\
&= \int_{\underline{r}}^1 (s - c_L) dF(s)^{N-1} - F(\beta^{-1}(\underline{r})) \int_{\underline{r}}^{\beta^{-1}(\underline{r})} (s - c_L) dF(s)^{N-1} - \int_{\beta^{-1}(\underline{r})}^1 F(v)(v - c_L) dF(v)^{N-1}. \quad (37)
\end{aligned}$$

**Derivative of the first term.**

$$\frac{d}{d\underline{r}} \int_{\underline{r}}^1 (s - c_L) dF(s)^{N-1} = -(N-1)(\underline{r} - c_L)F(\underline{r})^{N-2}f(\underline{r}). \quad (38)$$

**Derivative of the second term.**

$$\begin{aligned}
& \frac{d}{d\underline{r}} \left( F(\beta^{-1}(\underline{r})) \int_{\underline{r}}^{\beta^{-1}(\underline{r})} (s - c_L) dF(s)^{N-1} \right) \\
&= f(\beta^{-1}(\underline{r})) \frac{d\beta^{-1}(\underline{r})}{d\underline{r}} \int_{\underline{r}}^{\beta^{-1}(\underline{r})} (s - c_L) dF(s)^{N-1} \\
&\quad + (N-1)F(\beta^{-1}(\underline{r})) \left[ (\beta^{-1}(\underline{r}) - c_L)F(\beta^{-1}(\underline{r}))^{N-2}f(\beta^{-1}(\underline{r})) \frac{d\beta^{-1}(\underline{r})}{d\underline{r}} - (\underline{r} - c_L)F(\underline{r})^{N-2}f(\underline{r}) \right]. \quad (39)
\end{aligned}$$

**Derivative of the third term.**

$$\frac{d}{d\underline{r}} \int_{\beta^{-1}(\underline{r})}^1 F(v)(v - c_L) dF(v)^{N-1} = -\frac{d\beta^{-1}(\underline{r})}{d\underline{r}} (N-1)(\beta^{-1}(\underline{r}) - c_L)F(\beta^{-1}(\underline{r}))^{N-1}f(\beta^{-1}(\underline{r})). \quad (40)$$

The derivative of the double integral is therefore

$$-(N-1)(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r})(1-F(\beta^{-1}(\underline{r}))) - f(\beta^{-1}(\underline{r}))\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)dF(s)^{N-1}. \quad (41)$$

**Derivative of the bracket term.**

$$\begin{aligned} & \frac{d}{d\underline{r}} [(r-c_L)F(\underline{r})^{N-1}(1-F(\beta^{-1}(\underline{r})))] \\ &= \left[ F(\underline{r})(1-F(\beta^{-1}(\underline{r}))) \right. \\ & \quad \left. + (N-1)(\underline{r}-c_L)(1-F(\beta^{-1}(\underline{r})))f(\underline{r}) \right] F(\underline{r})^{N-2} \\ & \quad - (\underline{r}-c_L)F(\underline{r})f(\beta^{-1}(\underline{r}))\frac{1}{1-\delta\psi} \end{aligned} \quad (42)$$

### A.3 Full FOC and simplification

Taken together,  $d\Pi(\underline{r})/d\underline{r} = 0$  is

$$\begin{aligned} 0 &= \delta N(c_H - c_L)\frac{1}{1-\delta\psi}f\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-2} \\ & \quad \times \left[ NF\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right) - (N-1)F\left(\frac{c_H-\psi c_L}{1-\psi}\right) \right] \\ & \quad + N \left[ F(\underline{r})(1-F(\beta^{-1}(\underline{r}))) \right. \\ & \quad \left. + (N-1)(\underline{r}-c_L)(1-F(\beta^{-1}(\underline{r})))f(\underline{r}) \right] F(\underline{r})^{N-2} \\ & \quad - (\underline{r}-c_L)F(\underline{r})f(\beta^{-1}(\underline{r}))\frac{1}{1-\delta\psi} \\ & \quad + N \left[ -(N-1)(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r})(1-F(\beta^{-1}(\underline{r}))) \right. \\ & \quad \left. - f(\beta^{-1}(\underline{r}))\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)dF(s)^{N-1} \right]. \end{aligned} \quad (43)$$

The  $(N-1)(\underline{r}-c_L)$  terms from the bracket and double-integral derivatives cancel. Solving the remaining expression for  $\underline{r}-c_L$  gives (24). The outer factor  $N$  in the failure derivative and the outer factor  $N$  in the success derivative cancel in this final step.

## B Profit for $r > \underline{r}$

Write  $v_\psi \equiv (c_H - \psi c_L)/(1 - \psi)$  and  $v_{\tilde{\psi}}(v) \equiv ((1 - \psi\rho(\beta(v)))c_H - \psi(1 - \rho(\beta(v)))c_L)/(1 - \psi)$ . The stage-2 failure block in (25) is

$$\begin{aligned} & \delta N(c_H - c_L) (F(\beta^{-1}(\underline{r})) - F(v_\psi)) F(\beta^{-1}(\underline{r}))^{N-1} \\ & \quad + \delta N(c_H - c_L) \int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)} \left( 1 - \frac{F(v_{\tilde{\psi}}(v))}{F(v)} \right) dF(v)^N, \end{aligned} \quad (44)$$

plus the stage-1 success terms in (25). The first line matches (16) at  $r = \underline{r}$ . Since  $\int (1 - F_{\tilde{\psi}}/F) dF^N = N \int (F - F_{\tilde{\psi}}) F^{N-2} dF$ ,

$$\int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)} \left(1 - \frac{F_{\tilde{\psi}}}{F}\right) dF^N = F(\beta^{-1}(r))^N - F(\beta^{-1}(\underline{r}))^N - N \int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)} F_{\tilde{\psi}} F^{N-2} dF,$$

which yields the middle terms in (25).

## C Mixed-strategy indifference and the ODE

### C.1 Incremental indifference

On  $[\underline{r}, \bar{r}]$ , equilibrium requires  $\hat{\Pi}(r) = \Pi(\underline{r}^*)$ . The  $\psi$  block in (25) does not depend on  $r$ ; with  $\underline{r} = \underline{r}^*$  on path,

$$\begin{aligned} \delta N(c_H - c_L) & \left[ F(\beta^{-1}(r))^N - F(\beta^{-1}(\underline{r}^*))^N - N \int_{\beta^{-1}(\underline{r}^*)}^{\beta^{-1}(r)} F \left( \frac{(1 - \psi \rho(\beta(v)))c_H - \psi(1 - \rho(\beta(v)))c_L}{1 - \psi} \right) F(v)^{N-2} dF(v) \right] \\ & + N \left\{ (r - c_L) F(r)^{N-1} (1 - F(\beta^{-1}(r))) + \int_{\beta^{-1}(r)}^1 \int_r^v (s - c_L) dF(s)^{N-1} dF(v) \right\} \\ & = N \left\{ (\underline{r}^* - c_L) F(\underline{r}^*)^{N-1} (1 - F(\beta^{-1}(\underline{r}^*))) + \int_{\beta^{-1}(\underline{r}^*)}^1 \int_{\underline{r}^*}^v (s - c_L) dF(s)^{N-1} dF(v) \right\}. \end{aligned} \quad (45)$$

### C.2 Derivative and the ODE

On the low branch,  $\beta^{-1}(r) = h(r)$ . Differentiating (45) in  $r$  gives

$$\begin{aligned} 0 &= \delta N(c_H - c_L) \left[ F(h(r))^{N-1} f(h(r)) h'(r) - F \left( \frac{(1 - \psi \rho(r))c_H - \psi(1 - \rho(r))c_L}{1 - \psi} \right) F(h(r))^{N-2} f(h(r)) h'(r) \right] \\ & + F(r)^{N-1} (1 - F(h(r))) - (r - c_L) F(r)^{N-1} f(h(r)) h'(r) \\ & - f(h(r)) h'(r) \int_r^{h(r)} (s - c_L) dF(s)^{N-1}. \end{aligned} \quad (46)$$

Collecting terms with  $h'(r)$  in (46) and solving gives (29)–(30).

### C.3 Initial denominator

At the lower support,  $\rho(\underline{r}^*) = 0$ , so buyers keep the prior belief  $\psi$  after failure. The first bidding branch in (8) therefore gives

$$\beta^{-1}(\underline{r}^*) = \frac{\underline{r}^* - \delta \psi c_L}{1 - \delta \psi}.$$

The stage-2 cutoff in (30) also reduces to  $(c_H - \psi c_L)/(1 - \psi)$ . To evaluate the ODE denominator at the initial point, first use

$$\begin{aligned} & \int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)} (s - c_L) dF(s)^{N-1} \\ &= (\beta^{-1}(\underline{r}^*) - c_L) F(\beta^{-1}(\underline{r}^*))^{N-1} - (\underline{r}^* - c_L) F(\underline{r}^*)^{N-1} - \int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)} F(v)^{N-1} dv. \end{aligned}$$

Substituting this identity into (30) and using (24) gives

$$\begin{aligned} & \mathcal{B}(\underline{r}^*, \beta^{-1}(\underline{r}^*)) \\ &= (1 - \delta\psi) \frac{1 - F(\beta^{-1}(\underline{r}^*))}{f(\beta^{-1}(\underline{r}^*))} F(\underline{r}^*)^{N-1} \\ & \quad + \delta(c_H - c_L) F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) F(\beta^{-1}(\underline{r}^*))^{N-2}. \end{aligned}$$

This is the initial denominator used in Claim 3. It is strictly positive whenever  $F(\beta^{-1}(\underline{r}^*)) < 1$ .

#### C.4 High reserves

For  $r > \beta(v^*)$ ,  $\beta^{-1}(r) = (r - \delta c_H)/(1 - \delta)$  and (46) becomes

$$\begin{aligned} 0 = & \delta N(c_H - c_L) \left( F\left(\frac{r - \delta c_H}{1 - \delta}\right) - F\left(\frac{(1 - \psi\rho(r))c_H - \psi(1 - \rho(r))c_L}{1 - \psi}\right) \right) F\left(\frac{r - \delta c_H}{1 - \delta}\right)^{N-2} f\left(\frac{r - \delta c_H}{1 - \delta}\right) \frac{1}{1 - \delta} \\ & + F(r)^{N-1} \left( 1 - F\left(\frac{r - \delta c_H}{1 - \delta}\right) \right) \\ & - \frac{1}{1 - \delta} f\left(\frac{r - \delta c_H}{1 - \delta}\right) \left[ \left( \frac{r - \delta c_H}{1 - \delta} - c_L \right) F\left(\frac{r - \delta c_H}{1 - \delta}\right)^{N-1} - \int_r^{(r - \delta c_H)/(1 - \delta)} F(v)^{N-1} dv \right], \end{aligned} \tag{47}$$

which pins down  $\rho(r)$  on the upper part of the support.